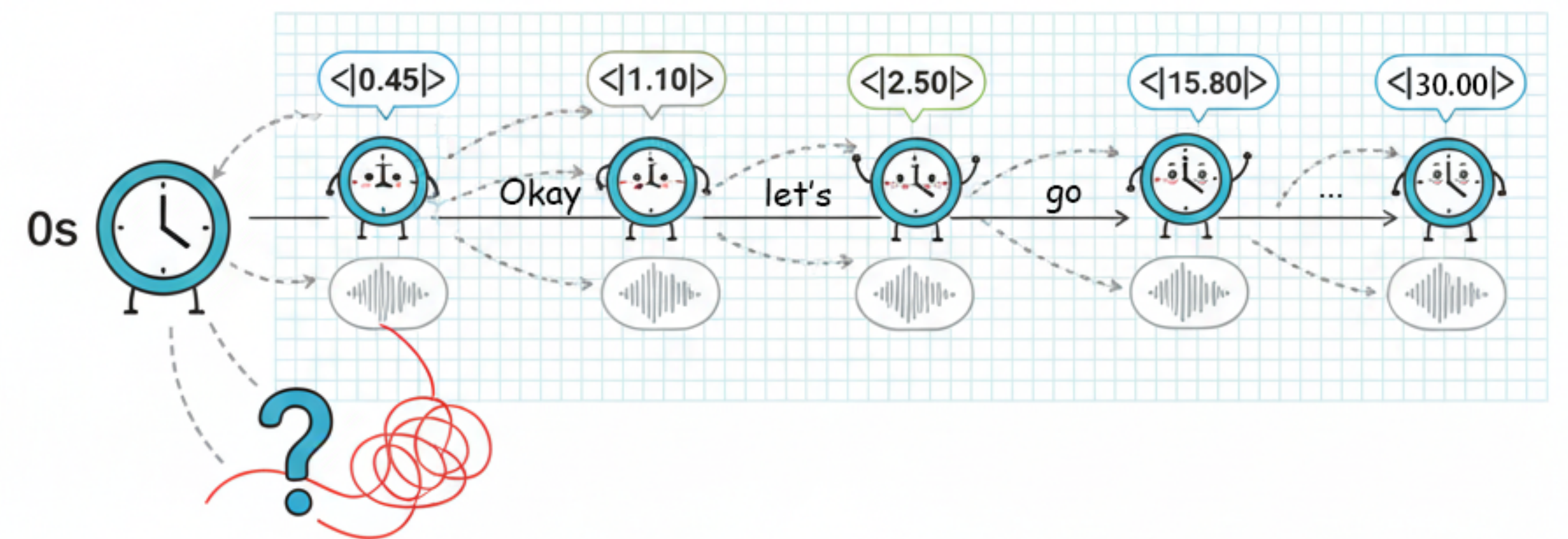


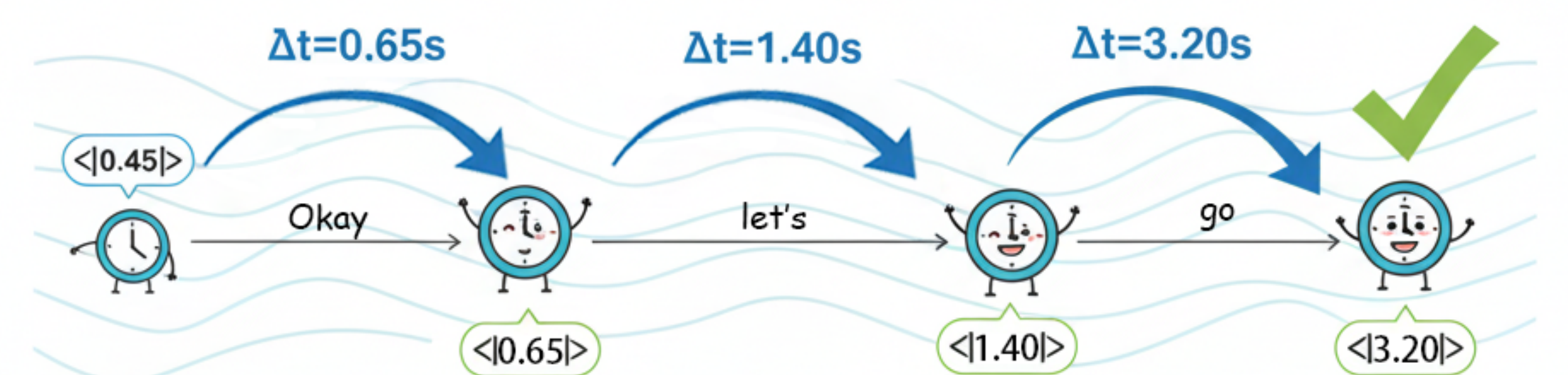
1. Motivation & Limitations

- **The Problem:** Speech LLMs excel at content understanding but struggle with fine-grained, word-level temporal alignment.
- **Absolute Timestamp Pitfalls:** Existing methods (e.g., Qwen2-Audio) use formats like $\langle [0.45] \rangle$ Okay $\langle [1.10] \rangle$ let's, leading to:
 - *Cumulative Error:* Monotonically increasing values cause error propagation.
 - *Poor Generalization:* Fails on audio longer than training duration (e.g., predicts $> 30s$ if only trained on 30s clips).
 - *Vocabulary Explosion:* 0.01s precision over 300s audio requires $\sim 30,000$ distinct timestamp tokens.
- **Our Solution: Relative Time Intervals & Masked Training**, transforming LLMs into “temporal-aware content understanding machines”. Instead of absolute positions, we model the **time interval** between consecutive words (e.g., $\langle [0.65] \rangle$ for words at 0.45s and 1.10s).
 - **Compact Vocabulary:** Only a limited range (e.g., 0.00s to 5.00s) is needed. Any length of audio can be represented cumulatively.
 - **Natural Paradigm:** Aligns perfectly with the autoregressive generation of LLMs. Avoids learning unrelated absolute values.

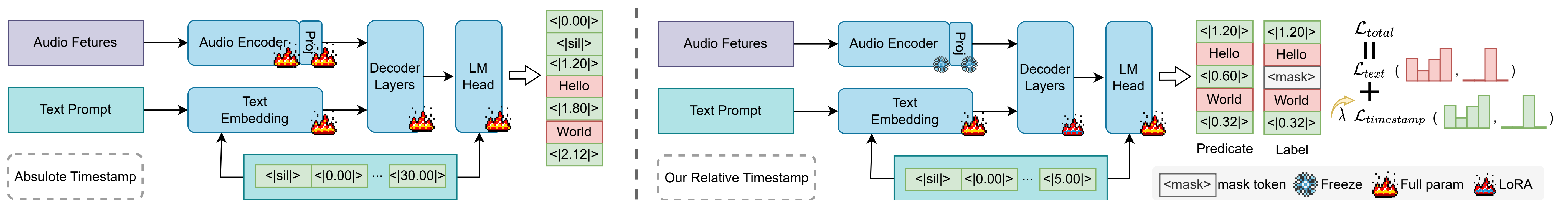
Traditional Absolute Timestmaps
(Fixed Reference)



Our Relative Timestmaps
(Dynamic Intervals)



2. Method Overview



3. Training Strategies & Fine-Tuning

1. Hybrid Parameter-Efficient Fine-Tuning:

- **Full-param:** Timestamp embedding layer & LM Head (learn new temporal vocabulary from scratch).
- **LoRA:** Decoder layers (preserve pre-trained linguistic knowledge, save compute).

2. Joint Temporal Alignment Loss:

$$L_{total} = L_{text} + \lambda L_{timestamp} \quad (1)$$

Dynamic weighting: λ starts at 1 (focus on text) and increments by 1 per epoch (shift focus to timestamps).

3. Timestamp Masking (Regularization):

- Randomly replaces previous timestamps with a $\langle \text{mask} \rangle$ token (10% probability from epoch 2).
- Prevents over-relying on ground-truth history. Forces the model to *listen to the actual audio pause* to infer time, enhancing robustness against noisy annotations.

4. Main Results

Our method sets a new SOTA for timestamp prediction while significantly improving ASR (WER).

Timestamp prediction (Wenet-Meeting) & ASR performance comparison.

Model	Tolerance (ms)	Precision (%)	Recall (%)	Avg. Diff (ms)	WER (%) (Wenet)
Qwen2-Audio	240	82.30	75.87	43.07	21.99
WhisperTS	240	43.97	18.79	102.48	56.66
SenseVoice	240	82.82	81.19	32.47	7.54
Abs. TS (Baseline)	240	72.41	68.22	152.17	11.30
Relative TS (Ours)	240	91.13	86.88	30.34	5.56

- Achieves **30.34 ms** average difference, the closest to ground truth.
- Surprisingly, adding the timestamp task **decreases WER** from 11.30% to 5.56%, proving temporal modeling forces better acoustic feature extraction.

6. Conclusion

We propose a novel framework addressing core challenges in precise temporal sequence understanding for Speech LLMs.

- **Beyond simple tagging:** Endows LLMs with fine-grained, aligned multi-modal spatiotemporal understanding capabilities.
- **Highly Scalable:** Compact vocabulary enables robust timestamping for *arbitrarily long audio* without degradation.
- **Mutual Enhancement:** Temporal alignment regularization actively improves standard ASR performance.

